Supplementary Material 2 – Full cross-correlation and windowed cross-correlation

The analysis technique we presented involves the computation of Pearson normalized cross-correlation curves. We compute regular full-length cross-correlations (CC). This technique allows us to explore global and persistent patterns that indicate similar motion in two people even if delayed in time. A different technique, windowed cross-correlation (WCC), is based on the same kind of computation but applies it to overlapping same-length windows at regular offsets (Boker et al., 2002).

This appendix briefly contrasts CC and WCC results with the same simple synthetic signals. We hope this clarifies the rationale behind our choice. A different choice could be valid depending on features that require focus, given the research question. It's important to make clear that the technique we presented is neither CC nor WCC, but an analytic procedure where one of the steps uses CC. We described that procedure, which uses CC, because it proved useful to find global patterns in human coordinated motion.

Figure {cc_pearson} shows synthetic input signal pairs in the left column, and CC vs WCC results in the right column, both based on Pearson's formula (that is, centered and normalized, which produces a correlation range between -1 and 1). For WCC, window size is 5 s, step is 0.5 s (90% overlap), and sample rate is 100 Hz. Input signal length is 20 s.

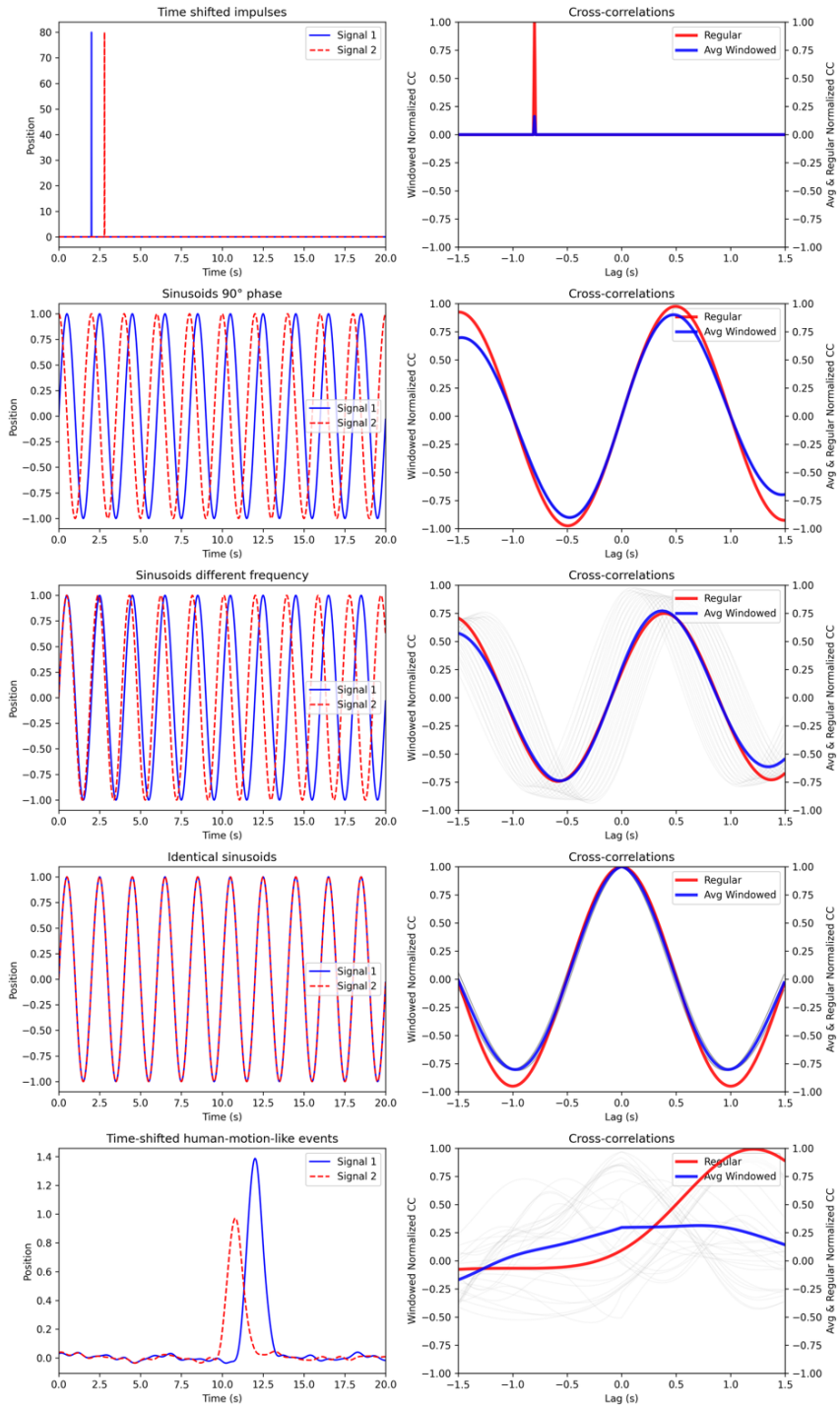
We observe that regular sinusoidal signals produce similar results. However, synthetic signals more related to motion, like an impulse or a wider peak, have a more present effect on the resulting cross-correlation curves. In this latter case, a curve computed from the average of WCC will reflect *coordinated* motion at a single or a few windows, and average it with the results from other windows, which will eventually show a weaker effect. Each window in the average has a little piece of temporal context. On the other hand, the computation of a single CC curve has the full context, so it will display that, at some lag, the two signals have essentially the same shape. That effect will not weaken by averaging with results from other windows of which there are none.

We can also see that the effect of a wider context is relevant even within a narrower lag result window. In the plotted examples, both CC (20 s) and WCC (5 s) operate on longer signals than the displayed lag window (-1.5 s to 1.5 s, total 3 s) for the resulting curves. Nevertheless, extending available context from 5 s to 20 s when going from WCC to CC still makes a difference in the 3 s lag range.

Figure {cc_speed} shows a similar analysis, but input signals are the first difference of signals in Figure {cc_pearson}. In other words, if original signals were position, these are the corresponding speed signals (units are irrelevant because of Pearson normalization). We can see that speed signals have the nice property of tending to make the *resting* level

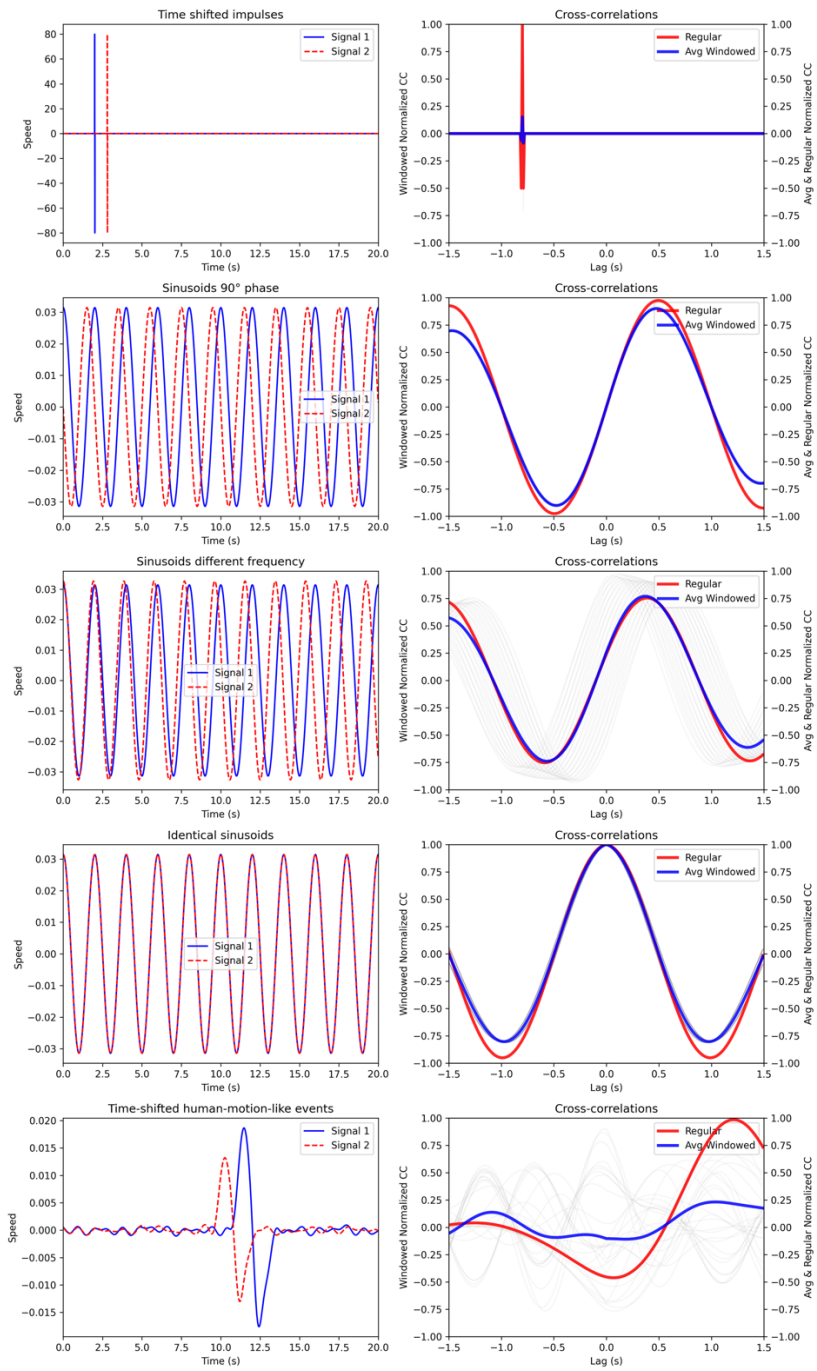
close to the vertical center. We still center signals when computing all cross-correlations in order to avoid any slight imbalance to show in the result. This high-pass-filter-like characteristic has a detrend effect that makes the full CC viable. Apart from that, previous comments on CC vs WCC for position signals are equally valid for speed signals.

Figure {cc_pearson}



Note. Left column shows pairs of synthetic position signals. Right column shows CC vs WCC results. For WCC, individual window cross-correlations are shown in light gray, sometimes covered by their average WCC curve. Sampling rate is 100 Hz. Input signal length is 20 s. WCC window size is 5 s, step is 0.5 s.

Figure {cc_speed}.



Note. Displays the same process as Figure {cc_pearson} but with different input signals. This time, inputs are the speed signals corresponding to position signals in Figure {cc_pearson}.